

```

import pandas as pd
import matplotlib.pyplot as plt

from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

%matplotlib inline

data = pd.read_csv("real_estate_maggie.csv")

print(data.shape)
data.head()

(120, 11)

```

	Property_ID	Area_sqft	Price_Lakh	Bedrooms	Age_years	\
0	P001	1776	173.1	5	27	
1	P002	2109	204.6	5	27	
2	P003	1510	108.5	4	4	
3	P004	1944	119.0	5	8	
4	P005	1780	119.2	4	3	

	Distance_to_City_km	Location_Score	Condition_Score	Amenities_Score \
0	19.8	5.0	3.3	
1	16.3	6.4	2.0	
2	20.3	5.7	9.5	
3	22.7	4.9	9.5	
4	15.8	6.3	10.0	

	Days_on_Market	Sold
0	133	Yes
1	180	Yes
2	32	Yes
3	50	Yes
4	25	Yes

```

unsold = data[(data["Days_on_Market"] > 180) & (data["Sold"] == "No")]

print("Number of properties unsold for more than 6 months:")
print(len(unsold))

unsold.head()

Number of properties unsold for more than 6 months:
2

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	Property_ID	Area_sqft	Price_Lakh	Bedrooms	Age_years	\
46	P047	2456	254.3	5	30	
116	P117	2422	252.4	5	30	

	Distance_to_City_km	Location_Score	Condition_Score	Amenities_Score	\
46	13.8	6.0	2.7	6.9	
116	17.7	6.1	5.5	9.9	

	Days_on_Market	Sold
46	181	No
116	195	No

```
features = [
    "Area_sqft",
    "Price_Lakh",
    "Bedrooms",
    "Age_years",
    "Distance_to_City_km",
    "Location_Score",
    "Condition_Score",
    "Amenities_Score",
    "Days_on_Market"
]
```

```
df = data[features]
```

```
print(df.head())
```

	Area_sqft	Price_Lakh	Bedrooms	Age_years	Distance_to_City_km	\
0	1776	173.1	5	27	19.8	
1	2109	204.6	5	27	16.3	
2	1510	108.5	4	4	20.3	
3	1944	119.0	5	8	22.7	
4	1780	119.2	4	3	15.8	

	Location_Score	Condition_Score	Amenities_Score	Days_on_Market
0	5.0	3.3	9.4	133
1	6.4	2.0	4.0	180
2	5.7	9.5	6.3	32
3	4.9	9.5	4.6	50
4	6.3	10.0	6.9	25

```
sc = StandardScaler()
```

```
scaled_data = sc.fit_transform(df)
```

```
principal = PCA(n_components=3)
```

```

principal.fit(scaled_data)

x = principal.transform(scaled_data)

print(x.shape)

(120, 3)

print("Principal Components:")
print(principal.components_)

Principal Components:
[[ 0.38822323  0.41522324  0.3437768   0.3108704   0.23322549 -
 0.24027007
  -0.32463858 -0.08302811  0.48748517]
 [ 0.44845108  0.35557202  0.44704566 -0.40139713 -0.20349941
  0.15407883
   0.42163731  0.05825859 -0.25527692]
 [ 0.0539468   0.06324605  0.04533305  0.36662173 -0.59172729
  0.60176013
  -0.31982558  0.20149859  0.03842626]]

print("Explained Variance Ratio:")
print(principal.explained_variance_ratio_)

print("Total variance explained by 3 components:")
print(principal.explained_variance_ratio_.sum())

Explained Variance Ratio:
[0.34383267 0.24397776 0.19950404]
Total variance explained by 3 components:
0.7873144722451758

plt.figure(figsize=(10, 7))

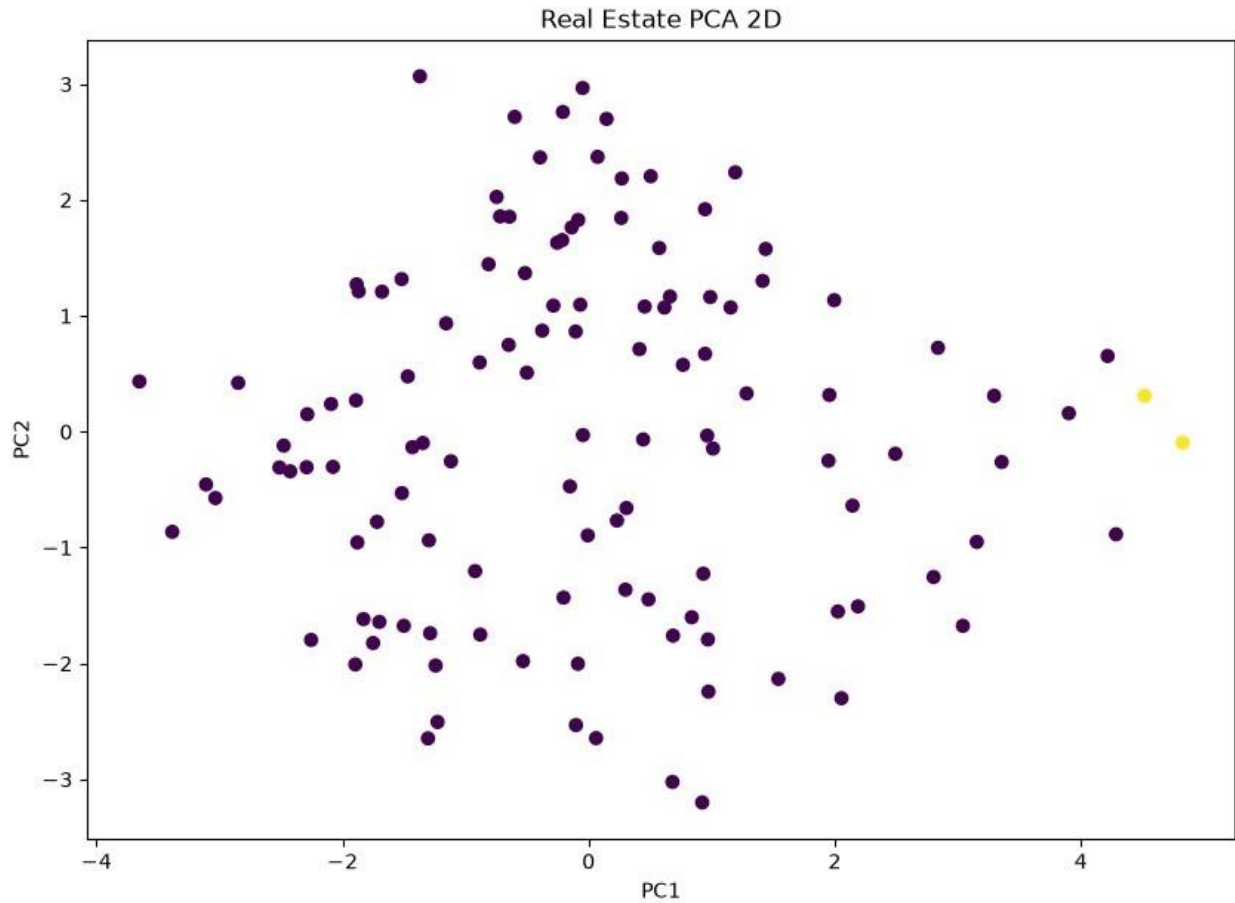
plt.title("Real Estate PCA 2D")

plt.scatter(
    x[:, 0],
    x[:, 1],
    c=(data["Sold"] == "No").astype(int)
)

plt.xlabel("PC1")
plt.ylabel("PC2")

plt.show()

```



```
loadings = pd.DataFrame(
    principal.components_.T,
    columns=["PC1", "PC2", "PC3"],
    index=features
)

print(loadings)
```

	PC1	PC2	PC3
Area_sqft	0.388223	0.448451	0.053947
Price_Lakh	0.415223	0.355572	0.063246
Bedrooms	0.343777	0.447046	0.045333
Age_years	0.310870	-0.401397	0.366622
Distance_to_City_km	0.233225	-0.203499	-0.591727
Location_Score	-0.240270	0.154079	0.601760
Condition_Score	-0.324639	0.421637	-0.319826
Amenities_Score	-0.083028	0.058259	0.201499
Days_on_Market	0.487485	-0.255277	0.038426

```
print("Average values of Sold properties:")
print(data[data["Sold"] == "Yes"][features].mean())
```

```

print("\nAverage values of Unsold properties:")
print(data[data["Sold"] == "No"][features].mean())

Average values of Sold properties:
Area_sqft      1614.677966
Price_Lakh      117.240678
Bedrooms        3.466102
Age_years       14.838983
Distance_to_City_km  13.011864
Location_Score   6.828814
Condition_Score   7.300000
Amenities_Score   6.686441
Days_on_Market   58.728814
dtype: float64

Average values of Unsold properties:
Area_sqft      2439.00
Price_Lakh      253.35
Bedrooms        5.00
Age_years       30.00
Distance_to_City_km  15.75
Location_Score   6.05
Condition_Score   4.10
Amenities_Score   8.40
Days_on_Market  188.00
dtype: float64

for pc in ["PC1", "PC2", "PC3"]:
    print("\n", pc)
    print(loadings[pc].abs().sort_values(ascending=False))

PC1
Days_on_Market      0.487485
Price_Lakh           0.415223
Area_sqft            0.388223
Bedrooms             0.343777
Condition_Score       0.324639
Age_years            0.310870
Location_Score        0.240270
Distance_to_City_km   0.233225
Amenities_Score       0.083028
Name: PC1, dtype: float64

PC2
Area_sqft            0.448451
Bedrooms             0.447046
Condition_Score       0.421637
Age_years            0.401397
Price_Lakh           0.355572

```

```
Days_on_Market      0.255277
Distance_to_City_km  0.203499
Location_Score       0.154079
Amenities_Score      0.058259
Name: PC2, dtype: float64
```

```
PC3
Location_Score      0.601760
Distance_to_City_km  0.591727
Age_years           0.366622
Condition_Score     0.319826
Amenities_Score     0.201499
Price_Lakh          0.063246
Area_sqft           0.053947
Bedrooms            0.045333
Days_on_Market      0.038426
Name: PC3, dtype: float64
```

INFERENCE:

PCA was applied to the real-estate dataset after standardizing the features using StandardScaler.

The principal components **help** identify the major factors that explain variation among the properties. Properties remaining unsold **for** more than six months can be compared **with** sold properties using price, location, condition, age, amenities **and** distance.

The PCA loadings **and** average values **help** identify the factors that have the strongest influence on the differences between properties.